Ans

1) $\sum n = \frac{n(n+1)}{2} = \frac{100 \times 101}{2} = 5050$

2) $1^3 + 2^3 + 3^3 + \ldots + 20^3$

$= \left(\frac{20 \times 21}{2}\right)^2 = 44100$

3) $11^3 + 12^3 + \ldots + 20^3$

$= (1^3 + \ldots + 20^3) - (1^3 + \ldots + 10^3)$

$= 40065$

4) $1^3 + 2^3 + \ldots + n^3 = 2025$

$\left(\frac{n(n+1)}{2}\right)^2 = 2025 \Rightarrow \frac{n(n+1)}{2} = 45$

$n = 9$

5) $(1 \times 2) + (2 \times 3) + (3 \times 4) + \ldots + n(n+1)$,

$n = 10$

$n(n+1)(n+2)/3 = 420$